

InterLeaf: A Letter-Based Mobile Platform for Reflection, Connection, and Chronic Pain Support

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Abstract

Chronic pain affects a large portion of the population and often leads to isolation. Many digital tools focus on symptom tracking, goal maintenance, or behavior monitoring, but they offer limited room for personal reflection or meaningful peer connection. This paper presents InterLeaf, a mobile app designed to support people living with chronic pain through reflective letter writing, comfort-item sharing, and intentional interpersonal connection. Our contribution is a calm, non-performative interaction design paired with an interpretable AI tagging pipeline used to (1) gate sharing based on potential harm signals and (2) surface editable tags that support browsing and resonance. We report an offline evaluation on a 250-letter synthetic corpus, focusing on safety detection, ontology tagging, free-form theme recovery, and confidence gating. Results suggest the pipeline can support cautious surfacing decisions when treated as a risk signal rather than a guarantee, and when user control and editability are central.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods; Health care information systems; User centered design; • **Computing methodologies** → Natural language processing.

Keywords

Chronic Pain, Community, Human-centered Design, Reflective Design, Digital Empathy, Narrative Sharing

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1 Introduction

Chronic pain affects about one in five Canadians [19] and remains a persistent challenge for both individuals and the healthcare system [4, 5]. Despite extensive research, there is still no reliable biomarker to confirm its presence and no definitive cure [14, 17]. As a result, its long-term nature often contributes to sustained feelings of loneliness and disconnection. Many individuals with chronic pain report isolation and difficulty maintaining social ties or staying connected to care networks [2, 7]. Recent studies also show a strong two-way correlation between loneliness and chronic pain [13]. This bidirectional link highlights the need for approaches that address the psychological impact of living with persistent pain.

Although digital health tools are widely used, most focus on symptom tracking, treatment adherence, or motivational goals. These systems prioritize measurement and self-monitoring, which leaves little room for personal expression, peer recognition, or the kinds of insight that come from lived experience [11]. This gap suggests the need for approaches that support connection rather than only measurement.

We introduce InterLeaf, a mobile application designed around three interaction layers: private letter writing, the sharing of meaningful objects in a communal “grove,” and selective one-to-one connection when users find resonance in each other’s narratives (see Fig. 1). InterLeaf extends reflective writing into an interactive context by incorporating an AI-driven semantic tagging system that identifies shared themes and experiences (see Fig. 2). The overall design follows principles of calm technology, remaining unobtrusive and avoiding social visibility metrics, with the goal of supporting emotional resonance rather than performance.

InterLeaf is designed to make reflective expression and small-scale peer connection easier to initiate and harder to perform for an audience. The goal is not treatment, diagnosis, or outcome improvement claims. This paper is about system design and whether an interpretable tagging pipeline can support cautious sharing and letter matching in a setting where the content is emotionally loaded.



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2 Related Work

Existing research lands in three clusters. Expressive writing is widely used to help people sort through difficult experiences [18], but it usually happens in isolation, with no one on the other end to witness or respond. In HCI, similar ideas appear in research on reflective systems, where the goal is to slow interaction and encourage users to examine their own assumptions rather than react automatically [10, 12], yet most existing digital reflection systems in this area are built around solitary use [22].

Many chronic pain communities rely on shared storytelling as a form of support, and platforms such as The Mighty [16] show that large numbers of people want spaces where they can talk openly about their conditions and read others' experiences, while youth-focused initiatives like Letters to Strangers [20] demonstrate that anonymous letter exchanges can create a sense of solidarity and mutual understanding. However, chronic pain communities rely on shared storytelling, yet public platforms distort those stories through visibility metrics (e.g., likes, comments, shares) and scale [1].

Work on digital empathy examines how systems can foster emotional presence without demanding rapid responses or exposing users to judgment [15], and calm technology offers a related perspective by advocating interfaces that stay in the periphery and avoid constant interruption [3, 6, 8]. Work on digital empathy and calm technology pushes back on that pace, arguing for systems that support emotional presence without pressure [21]. InterLeaf responds to these failure modes by keeping reflection private-first, making sharing optional and non-ranked, and constraining connection to opt-in 1:1 exchanges rather than broadcast threads.

3 System Design

3.1 Interaction Layers

InterLeaf consists of three interaction layers (Fig. 1) designed to reduce audience pressure and keep disclosure under user control.

Letters. Users write private letters that can remain private or be shared anonymously. There are no likes, comments, or follower metrics, because InterLeaf is trying to avoid the “post for a reaction” loop. Sharing is a deliberate choice rather than an expectation.

Grove of Solace. The Grove is a shared gallery where users post comfort objects with short captions. Items remain available over time rather than being pushed down by frequent updates, so participation is not a race for attention.

Chats. Connections start when someone reads a shared letter and chooses to reach out. Conversations are asynchronous and 1:1 by default, which keeps contact intentional and reduces the blast radius of misinterpretation.

4 AI Tagging Pipeline

The AI pipeline (Fig. 2) produces two outputs: (1) a safety risk signal used to gate sharing and matching, and (2) a small set of editable tags used for browsing and lightweight resonance. It includes narrative normalization, harm screening, contextual tag generation, ontology-based keyword tagging, similarity matching, and a final fusion step with confidence gating.

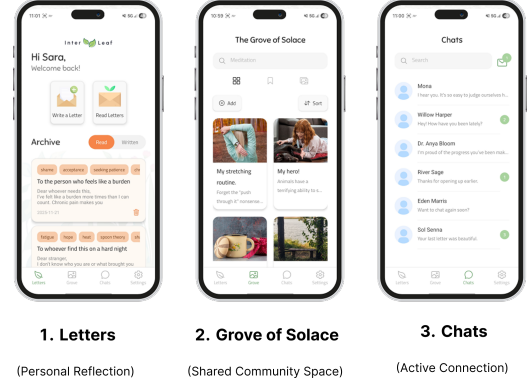


Figure 1: InterLeaf 3 main interaction layers: 1) Letters, 2) Grove of Solace, and 3) Chats.

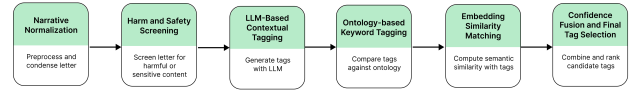


Figure 2: AI Tagging Pipeline.

Table 1: Ontology categories and canonical term labels.

Category	Canonical labels
Body area	head, neck, shoulder, back, hip, knee, hands, feet, diffuse
Symptom	burning, stabbing, throbbing, stiffness, fatigue, brain fog
Trigger	weather, stress, long sitting, hormonal cycle
Coping	pacing, heat, mindfulness, medication, stretching
Emotion	grief, anger, frustration, shame, hope, loneliness

The ontology specifies five conceptual categories: body areas, symptoms, triggers, coping strategies, and emotions as listed in Table 1. If no ontology term meets the similarity threshold, the candidate retains its original free-form label. The system generally assigns three to five tags per letter.

5 Methodology

Before using the AI tagging pipeline with real participants, we evaluated it offline on a synthetic but task-relevant corpus. The idea was to exercise the system on letters that look and feel like InterLeaf writing as a feasibility check before collecting real user letters. The dataset consists of 250 letters designed to approximate the anticipated writing style in InterLeaf. We generated these letters with a large language model (ChatGPT), prompted with short descriptions of chronic pain situations, emotional tones, and coping profiles.

Each letter was annotated along three layers: safety, ontology tags, and free-form tags. Two trained human annotators carried out all labeling using a shared codebook that defined each label and provided examples for common and borderline cases. Both annotators independently labeled all 250 letters with safety status, ontology tags across the five domains, and up to two free-form

Table 2: Safety confusion matrix.

	Harmful	Not harmful	Total
Harmful	38	15	53
Not harmful	4	193	197

tags as needed. Once disagreements were resolved, we treated the consensus labels as the reference standard for all evaluations.

The pipeline was evaluated on four tasks: **binary safety detection** (can it separate harmful vs not harmful letters by thresholding its safety score and measuring precision/recall/F1), **ontology multi-label tagging** (does it match human-selected tags across the ontology, using set-overlap metrics like precision/recall/F1 and Jaccard similarity), **free-form theme recovery** (does it surface important out-of-ontology themes both in candidate tags and final fused tags), and **confidence calibration** (do higher confidence scores actually correspond to higher accuracy, and do they track human certainty).

6 Results

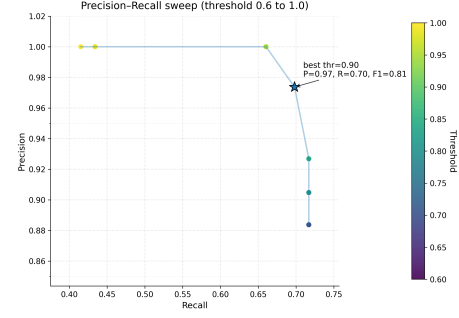
All results use consensus labels and high-confidence human annotations (confidence ≥ 2) and address RQ1–RQ4:

(RQ1 Safety) In 250 letters (53 harmful, 197 not harmful), the default threshold 0.80 yielded precision = 0.905, recall = 0.717, $F1 = 0.800$ for harmful, and accuracy = 0.924 (TP= 38, FN= 15, FP= 4, TN= 193); false negatives typically mixed chronic despair with indirect self-harm language (e.g., “sometimes I wish I could disappear”), while false positives used extreme distress without explicit self-harm intent; the best- $F1$ point was at threshold 0.9 with precision = 0.97, recall = 0.70, $F1 = 0.81$, accuracy = 0.93 (precision–recall trade-off).

(RQ2 Ontology tagging) Across all domains, macro precision = 0.797, macro recall = 0.787, macro $F1 = 0.766$, and micro $F1 = 0.769$, with micro-level counts TP= 1494, FN= 414, FP= 486, TN= 8106; per-letter overlap had mean Jaccard = 0.633 (SD = 0.143); by domain, symptoms and emotions were strongest (macro $F1 = 0.826$ and 0.803), triggers and coping were moderate (0.787 and 0.762), and body areas were weakest (0.660) due to under-detecting *diffuse* and over-predicting specific locations (e.g., *shoulder*, *hands*).

(RQ3 Free-form recovery) Free-form themes appeared in 102/250 letters (40.8%); with cosine threshold 0.6 on letters with at least one high-confidence free-form theme ($n = 102$), any-hit recall was 0.559 at the candidate stage (up to 15 candidates) and 0.588 after final output, while final recall@ K was macro/micro 0.451/0.452 at $K = 10$ and 0.387/0.376 at $K = 5$, i.e., fewer than half of gold free-form themes are recovered on average.

(RQ4 Confidence) For ontology tags, percentile bands ($p_{25} = 0.695$, $p_{75} = 0.873$; $n = 3612$) gave correctness 95.5% (high), 34.6% (medium), and 0.7% (low), implying confidence is useful as a high-end gate rather than a smoothly calibrated scale; safety confidence was saturated (quartile cutoffs both 1.0), but rank-based quartiles still showed accuracy 100.0% (high), 99.2% (medium), and 69.4% (low) over $n = 250$ letters; model vs human confidence correlated moderately (Spearman $\rho = 0.550$).

**Figure 3: Precision-recall curve across thresholds (harmful class).****Table 3: Ontology-tagging performance by domain.**

Domain	Macro P	Macro R	Macro F1
Body areas	0.740	0.720	0.660
Symptoms	0.839	0.826	0.826
Triggers	0.790	0.834	0.787
Coping	0.766	0.786	0.762
Emotions	0.853	0.778	0.803

Table 4: Ontology tag correctness by confidence band.

Band	Correctness	Comment
High	95.5%	Reliable
Medium	34.6%	Weak
Low	0.7%	Unreliable

7 Discussion

Overall, the safety signal is useful for adding friction and reducing exposure, but recall of 0.717 implies a meaningful miss rate, so it should be treated as a risk signal rather than a standalone safeguard for sharing and matching. Ontology tagging is generally consistent and most dependable for symptoms and emotions, while triggers and coping depend more on phrasing and context. Body-area tagging is the weakest part of the ontology and tends to introduce incorrect specificity, so those tags should be treated as editable suggestions and the system should explicitly support global or diffuse pain descriptions. Free-form theme recovery is too inconsistent to drive matching on its own, implying that contextual themes need separate handling (e.g., distinct presentation or a dedicated matching channel) if they are important to the product. Finally, confidence scores are useful mainly for gating: high-confidence outputs can be shown as primary descriptors, medium-confidence outputs should be framed as tentative, and low-confidence outputs should be suppressed by default unless a user opts in.

8 Limitations

This study is intentionally narrow and should be interpreted as an offline evaluation, not evidence of deployment robustness. The main limitation is the corpus: 250 gold-labeled letters, but not drawn

from a live InterLeaf population, and synthetic/curated text is typically more explicit and coherent than real writing produced under pain, fatigue, or emotional strain. Real letters will likely include typos, shorthand, fragmented narratives, code-switching, and implicit references, so reported performance may be optimistic. A second risk is distributional coupling: model-generated text evaluated by another model family can inflate results by compressing ambiguity and encouraging canonical phrasing [9]; stronger evidence requires real letters from people with lived experience and explicit measurement of the synthetic-to-real generalization gap.

The annotation setup is rigorous but too small for reliable subgroup analysis, and the ontology reflects an English-language, Western framing with unknown cultural and linguistic transfer. Several pipeline components were not evaluated in isolation (normalization, de-identification fidelity, evidence-span faithfulness), limiting claims about privacy-preserving meaning retention and explanation quality. Finally, we did not test downstream outcomes (matching quality, user experience, or fairness), so potential differential failures across writing styles or groups remain unmeasured. Next steps are to replicate on real pilot letters with consent, expand and co-design the ontology (especially coping and contextual themes), refine fusion/confidence to avoid overconfident errors and preserve context, and run end-to-end user studies plus targeted fairness audits based on realistic usage patterns.

InterLeaf handles sensitive narrative data (letters, chat messages, and comfort-item captions) plus derived metadata (tags and safety scores). This paper does not claim regulatory compliance or readiness for clinical deployment. If deployed beyond a research prototype, the system would require a clear data retention policy, secure storage and access controls, deletion/export support, and a defined response policy for high-risk content.

9 Conclusion

InterLeaf is a letter-based platform for people living with chronic pain that supports reflective writing, comfort-item sharing, and small-scale connections. It pairs a calm, non-performative interaction model with an AI tagging pipeline that extracts semantically grounded metadata from narratives so letters can resonate with others. The goal is not to measure or grade people; tags are meant to scaffold reflection and connection, not define feelings or prescribe coping.

An offline evaluation on a synthetic but task-aligned corpus suggests the approach is viable within clear limits. The safety module can serve as a risk signal to add friction and suppress sharing/matching under uncertainty, but recall limits mean it cannot be treated as a standalone safeguard. Ontology tagging largely tracks human judgments for explicit body areas, symptom types, and common negative emotions, and the contextual stage often notices themes beyond the ontology. However, weaknesses are design-relevant: subtle body-area tags are missed, contextual themes are often lost during fusion, and performance likely looks better here than it will on real user writing.

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